

What is the clinical significance of ABL1 E453V in Chronic Myeloid Leukemia with Dasatinib in CIViC?

Sure! I'll research the clinical significance of the ABL1 E453V mutation in Chronic Myeloid Leukemia, specifically in the context of treatment with Dasatinib, based on data from the CIViC database. I'll share my findings and provide the relevant citations.

Worked for 2 minutes >

CIViC contains a single evidence record (EID4720) for the **ABL1 E453V** missense mutation in *Chronic Myeloid Leukemia* (CML) treated with **dasatinib**. This evidence item is considered *predictive* and stems from a retrospective study of 1 084 chronic-phase CML patients who were resistant or intolerant to imatinib. Patients with a mutation in the ABL1 portion of the BCR-ABL fusion, which includes the E453V substitution, achieved **favourable responses to dasatinib monotherapy**—complete haematologic response (CHR) was 92 % (362 of 402 patients), major cytogenetic response (MCyR) was 56 % (225 of 402), complete cytogenetic response (CCyR) was 44 %, and major molecular response (MMR) was 39 % (125 of 383) . These response rates were comparable to or better than those in patients without an ABL1 mutation .

In CIViC the evidence item is classified as **Level C (case study)** with a *somatic* variant origin. The clinical significance is described as “**Sensitivity / Response**” to dasatinib therapy . Thus, according to CIViC, **ABL1 E453V predicts sensitivity to dasatinib in chronic myeloid leukemia**, supporting its use as a potentially effective therapy for patients harbouring this mutation.

Sources

Describe a task

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